Smart Docklands Proposal – Smart Bike Rental Management System

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GitHub repository: https://github.com/meganb01/BikeRentalApp

The theme chosen for this project is Mobility Innovation, one of the key focus areas in the Smart Docklands Pilot Call. Mobility innovation aims to explore new and sustainable ways to improve transportation in urban areas while ensuring accessibility, efficiency, and environmental sustainability. In busy urban environments such as Dublin Docklands, encouraging sustainable travel options such as cycling can help reduce congestion, lower carbon emissions, and promote healthier lifestyles. However, shared mobility systems such as bike rental services often face operational challenges, including inefficient bike distribution, lack of available bikes at peak times, and difficulties managing user demand. This project offers a solution that shows how these challenges could be addressed using efficient data structures.

This project proposes a Smart Bike Rental Management System designed to simulate how a bike sharing service could manage bikes and users efficiently. The application allows operators to track bikes in the system, manage users waiting to rent bikes, and monitor returned bikes. By using appropriate data structures, the system demonstrates how digital tools can improve mobility services and make transportation systems more efficient.

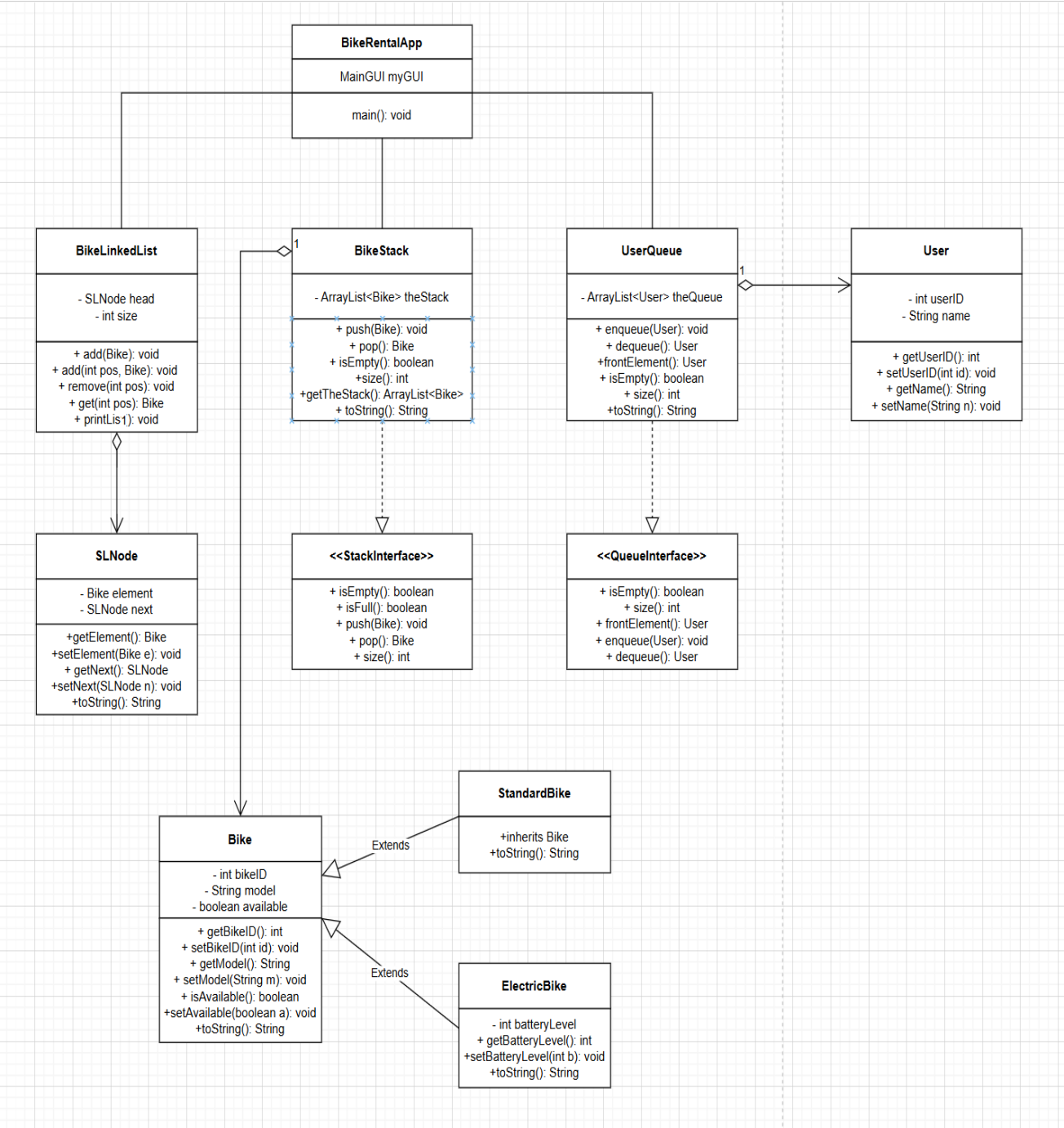
The system manages three main types of data: bikes, users waiting to rent bikes, and returned bikes. A Singly Linked List will be used to store all bikes in the system. This structure allows bikes to be dynamically added, removed, updated, and displayed, demonstrating the CRUD operations required by the application. Using a linked list allows the system to efficiently manage a flexible number of bikes without needing a fixed size data structure.

A Queue data structure will be used to manage users waiting to rent bikes. Queues follow a First-In-First-Out principle, which ensures that users are served in the order in which they arrive. When a bike becomes available, the user at the front of the queue will be assigned the bike first. This accurately models how waiting lines typically operate in real-world systems and ensures fairness in allocating shared resources.

A Stack data structure will be used to store recently returned bikes. Stacks follow a Last-In-First-Out principle, meaning the most recently returned bike can be accessed quickly. This can simulate scenarios where recently returned bikes can be quickly reassigned to new users.

The system will be implemented using Java NetBeans, with a Swing GUI that allows the user to interact with the application. The GUI will include forms for adding bikes, removing bikes, updating bike information, and displaying all bikes currently in the system. Additional controls will allow users to request a bike, return a bike, and view the queue of waiting users.

The project will also include inheritance and polymorphism through different types of bikes, such as standard bikes and electric bikes, which will extend a Bike class. Interfaces will also be implemented to define the behaviour of the Queue and Stack abstract data types.

Overall, this application demonstrates how core data structures can support the development of efficient mobility management systems. By simulating a bike sharing service for the Docklands area, the project aligns with the Mobility Innovation theme by exploring how digital tools and structured data management can contribute to smarter and more sustainable urban transportation systems.

Class Diagram: